Tasks:

  - Write a custom planner using prompt engineering strategies from section 11.3, ensuring it supports sequential task execution.
  - Integrate this planner into Nexus, and create an agent that uses it.
  - Test the planner with a complex goal that involves multiple steps and tools (e.g., data retrieval, processing, and saving).
  - Evaluate how the custom planner performs compared to built-in planners in Nexus or other platforms.
- *Exercise 5*—Implement Error Handling and Feedback in Sequential Planning (Advanced)

Objective —Learn how to implement error handling and feedback to refine sequential planning in an agentic system.

Tasks:

  - Using a sequential planner, set up an agent to perform a goal that may encounter common errors (e.g., a failed API call, missing data, or invalid input).
  - Implement error-handling mechanisms in the planner to recognize and respond to these errors.
  - Add feedback loops to adjust the plan or retry actions based on the error encountered.
  - Test the system by deliberately causing errors during execution, and observe how the agent recovers or adjusts its plan.

Summary

- Planning is central to agents and assistants, allowing them to take a goal, break it into steps, and execute them. Without planning, agents are reduced to simple chatbot-like interactions.
- Agents must differentiate between parallel and sequential actions. Many LLMs can handle parallel actions, but only advanced models support sequential planning, critical for complex task completion.